23-S2-Q4

(a) 4 clustering methods + application

(b) HAC : single.

Solution (a)

① (1) Centroids-based clustering

(2) Describe:

Centroids-based clustering is also known as paritifitioning method. It is a type of clustering that divides the data into non-hierachical groups. The most common example of paritifitioning clustering is the k-means clustering algorithm.

(3) Senario: we roughly know how many clusters (k) we want and the dusters are roughly spherical in feature space.

② (1) Hierachical clustering

(2) Hierachical clustering is one of the most commonly used methods for summarizing the property of data structure. A hierachical tree is a nested set of partition of samples, represented by a tree diagram or dendrogram

(3) Senario : we don't need to know the number of clusters and want a multi-level view of the clustering structure.

③(1) Distribution-based clustering

(2) The data is divided based on the probability of how a dataset belongs to a particular distribution.

The grouping is done by assuming some distributions, e.g. Gaussian distribution. The example of this type is the Gaussian Mixture Models (GMM).

(3) Scenario

when data are well-modeled by parametric distributions, or when we want a probabilistic interpretation

④(1) Density-based clustering

(2) The Density-based clustering method connects the highly dense areas into clusters, and arbitrarity shaped distributions are formed as long as the dense region can be connected.

(3) Scenario: When clusters are of arbitrary shape and you also want to identify outliers / noise. Common in spatial data, image analysis, anomaly detection.

(b) ①

$d_{s_1,s_2} = 0.8137$

$d_{s_1,s_3} = 1.7223$

$d_{s_1,s_4} = 2.1423$

$d_{s_2,s_3} = 2.3113$

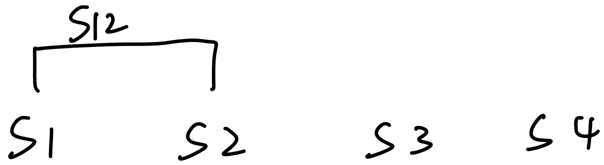
$d_{s_2,s_4} = 2.4234$

$d_{s_3,s_4} = 2.4387$

	S_1	S_2	S_3	S_4
S_1	0			
S_2	0.8137	0		
S_3	1.7223	2.3113	0	
S_4	2.1423	2.4234	2.4387	0

d_{s_1,s_2} is smallest, so we merge them and denote it as S_{12}

The dendrogram is as follow



② $d_{S12,S3} = \min\{d_{S1,S3}, d_{S2,S3}\}$

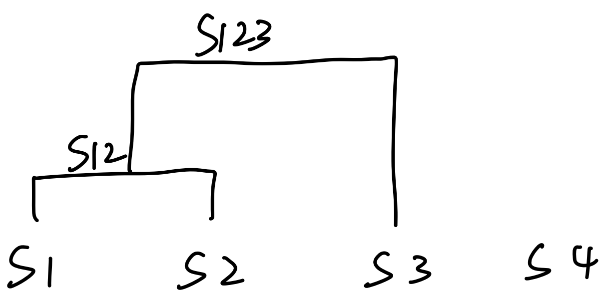
$= 1.7223$

$d_{S12,S4} = \min\{d_{S1,S4}, d_{S2,S4}\}$

$= 2.1423$

	S12	S3	S4
S12	0		
S3	1.7223	0	
S4	2.1423	2.4387	0

$\{S12, S3\} \to S123$



③ $d_{S123, S4} = \min\{d_{S1,S4}, d_{S2,S4}, d_{S3,S4}\}$

$= 2.1423$

	S123	S4
S123	0	
S4	2.1423	0

$\{S123, S4\} \rightarrow S1234$